

TASK-EFFICIENT AGENT EVALUATION USING CACHED TRACES

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ABSTRACT

Efficient benchmarking estimates a system’s full-benchmark performance from a small number of task executions. Standard approaches rely solely on the binary correctness of executions from a collection of already-evaluated reference systems. An agentic system, however, produces a rich trace for each execution that can reveal substantially more about its capabilities. Exploiting this information requires useful trace representations. We propose representing a trace by extracting a structured, rubric-guided summary of it with an LLM and embedding the summary with a standard text encoder. These representations preserve correctness- and task-related signal while minimizing the presence of system identity-, model family-, and harness-related signals that dominate raw trace embeddings. Given this representation for each task execution from all reference systems, we construct low-dimensional representations of the systems, predict a new system’s score from its position among the reference systems, and combine the resulting geometric prediction with a two-parameter logistic item-response model. On SWE-bench Verified and Terminal-Bench 2.0, the proposed estimator reduces mean absolute error from 0.13 to 0.08 ($p < 10^{-4}$) and from 0.14 to 0.12 ($p < 10^{-3}$), respectively, when provided a single random executed task relative to an item-response model alone. These results indicate that trace geometry and correctness outcomes carry complementary predictive signal and enable more accurate benchmark score estimation from limited agent executions.

1 INTRODUCTION

The dominant paradigm of generative model use is increasingly agentic: systems plan, invoke tools, observe intermediate results, and iterate over long horizons to complete complex tasks (Yang et al., 2024; Wang et al., 2025). As agentic capabilities increase, so does the range of settings in which a given system might be deployed – and, with it, the number of benchmarks on which the system must be evaluated before it can be trusted with consequential work (Kapoor et al., 2026). Agentic evaluation, however, is expensive as each task execution requires a properly sandboxed environment and a single task execution routinely consumes hundreds of thousands of tokens (Jimenez et al., 2024; Terminal-Bench Team, 2026) and can regularly exceed \$2,000 (Kapoor et al., 2026). Given this cost, efficient evaluation is necessary to increase the evaluation coverage of an agentic system.

Task-efficient evaluation methods (e.g., benchmark compression and anchor-item selection (Vivek et al., 2024; Maia Polo et al., 2024), item response theory and adaptive testing (Kipnis et al., 2025; Truong et al., 2025), leveraging cached responses (Helm et al., 2026a), etc.) address this problem by finding structure in item-level information available from previously-evaluated reference systems to decrease the cost of evaluating a new system. Some task-efficient evaluation methods require only the item-level scores from reference systems (Vivek et al., 2024; Kipnis et al., 2025). Others leverage the content of cached artifacts, such as natural language responses or residual streams, to measure the similarity of systems on a set of tasks (Duderstadt et al., 2023; Helm et al., 2026b;a). Notably, agentic workflows often produce artifacts when executing a task – file diffs, tool calls, intermediate plans, and error messages – recorded together in an execution trace that can be used to this effect. Unfortunately, existing methods for using cached artifacts rely on off-the-shelf embedding functions that are not immediately well-suited for representing traces.

We make two primary contributions:

Table 1: Mean Absolute Error (MAE) of benchmark score estimators for SWE-bench Verified and Terminal-Bench 2.0 under leave-one-family-out evaluation and different task selection protocols (random & adaptive). **Bolded** values are the best method for a given benchmark, task selection, and number of tasks. Underlined values are statistically significant ($\alpha = 0.05$) comparisons to IRT per the two-sided Wilcoxon signed-rank test. The TRUBRIC and IRT ensemble – which takes advantage of both trace geometry and per-execution score – is the best or near-best in all settings.

Num. tasks m	SWE-bench Verified						Terminal-Bench 2.0					
	random			adaptive			random			adaptive		
	1	5	20	1	5	20	1	5	20	1	5	20
sample score	0.442	0.175	0.082	0.380	0.151	0.072	0.396	0.164	0.073	0.360	0.124	0.132
IRT	0.127	0.088	0.048	0.080	0.062	0.034	0.137	0.097	0.052	0.112	0.067	0.039
raw trace	<u>0.099</u>	0.093	0.089	0.096	0.083	0.081	0.135	0.126	0.120	0.128	0.125	0.113
generic	0.079	<u>0.071</u>	0.068	0.074	0.070	0.069	0.139	0.127	0.115	0.144	0.130	0.117
trubric	<u>0.081</u>	<u>0.069</u>	0.060	0.068	0.056	0.060	<u>0.127</u>	0.107	0.093	0.110	0.092	0.084
IRT + trubric	<u>0.081</u>	0.063	0.043	0.062	0.048	0.034	0.122	0.090	0.050	0.099	0.059	0.037

1. **TRUBRIC trace representations.** We introduce TRUBRIC (*trace rubric*): a trace representation constructed by extracting a task specific structured summary of each trace with an LLM and embedding the summary with a standard text encoder. We show that TRUBRIC embeddings preserve information relevant to the task and behavior of the system while minimizing system-, model family-, and harness- specific signal across seven modern text encoders.

2. **Task-efficient evaluation using cached traces.** Given the TRUBRIC embeddings from a set of tasks for a target system and a collection of reference systems, we evaluate their utility for predicting the target system’s full-benchmark score on SWE-bench Verified and Terminal-Bench 2.0 under three different agentic system evaluation protocols. We find that TRUBRIC-based estimators, combined with an item-response model, reduce single-task mean absolute error from 0.13 to 0.08 ($p < 10^{-4}$) and from 0.14 to 0.12 ($p < 10^{-3}$) on the two benchmarks, respectively, and are nearly always better than baselines at any task budget (Table 1).

1.1 PROBLEM STATEMENT

For our purposes, an agentic system $f \in \mathcal{F}$ is a random mapping from a task space $\mathcal{Q}$ to a trace space $\mathcal{X}$. A benchmark is a fixed collection of tasks $Q^* = \{q_1, \dots, q_M\} \subset \mathcal{Q}$. Each trace is evaluated by the benchmark’s trace-level scoring function $s : \mathcal{X} \rightarrow \{0, 1\}$ and the system is evaluated by its benchmark score $y(f) = \frac{1}{M} \sum_{j=1}^M \mathbb{E}[s(f(q_j))]$ where the expectation is with respect to the randomness of f . We write y for $y(f)$ when the context is clear.

When evaluating a new system, we typically have access to $\mathcal{D} = \{\{(x_{ij}, y_{ij})\}_{j=1}^M\}_{i=1}^n$, the traces and trace-level scores of n previously-evaluated reference systems $f_1, \dots, f_n$, where $x_{ij} \sim f_i(q_j)$, $y_{ij} = s(x_{ij})$ and $y_i = y(f_i)$. For the target system f_0 , we observe traces (and, optionally, trace-level scores) on only a subset $Q \subset Q^*$ of $|Q| = m \ll M$ tasks and our goal is to estimate the system’s full benchmark score y_0 using only the target’s m executions and $\mathcal{D}$.

2 RELATED WORK

Representing agentic traces. The standard tools for embedding model outputs are text encoders trained on sentence-level semantic similarity (Reimers & Gurevych, 2019; Su et al., 2023; Nussbaum & Duderstadt, 2025). Agentic traces, however, implicitly or explicitly encode system metadata that may not be relevant for downstream tasks (Sun et al., 2025; Oderinwale, 2026) and are not necessarily encoded well by semantic embedding models. Instead, traces have primarily been studied through taxonomies of failure modes and actions (Cemri et al., 2025; Shu et al., 2026) and LLM judges of agent behavior (Zhuge et al., 2025; Mehtiyev & Assunção, 2026). Parallel work uses LLMs to adapt model outputs to be more naturally compared via their semantics (Ravfogel et al., 2024; Peng et al., 2024; Benara et al., 2024). TRUBRIC applies this to agentic traces by embedding a task-specific structured summary of each trace, preserving behavior-relevant information while minimizing the presence of system metadata.

Efficient benchmarking. Task-efficient benchmarking estimates a system’s full benchmark score from a small number of executions. The dominant paradigm for efficient benchmarking centers around item response theory (IRT) (Birnbbaum, 1968; Embretson & Reise, 2000), where a target system’s score is estimated with $m \leq M$ item-level scores by assigning a latent ability estimated from the relationship between a task’s difficulty and discriminability based on $\mathcal{D}$ (Maia Polo et al., 2024; Kipnis et al., 2025). Extensions of IRT select informative or adaptive task subsets (Vivek et al., 2024; Truong et al., 2025) or apply it to agentic benchmarks directly (Ge et al., 2026; Ndzomga, 2026). A complementary paradigm represents the systems themselves, either from their score profiles (Ruan et al., 2024; Zhuang et al., 2025) or from the artifacts they produce while executing a task (Duderstadt et al., 2023; Acharyya et al., 2024; Helm et al., 2025). For example, the Product Kernel Perspective Space (PKPS) (Anonymous, 2026) represents language models as Euclidean objects obtained via multidimensional scaling of the distances between their embedded responses. Helm et al. (2026a) regress benchmark scores on similar representations and prove that the resulting predictions are task-efficient relative to the sample score. TRUBRIC enables extending geometry-based methods to agentic traces by providing an effective trace encoding. We show that these two approaches – score-based IRT and artifact-based PKPS – can be combined to outperform either for any number of given executed tasks.

3 METHODS

We first present TRUBRIC and compare it to baseline trace embedding functions. We then describe constructing representations of the systems $f_0, f_1, \dots, f_n$ based on TRUBRIC embeddings that can be used to train a simple regressor in $\mathbb{R}^d$ for efficient benchmarking.

3.1 TASK-SPECIFIC RUBRICS FOR REPRESENTING TRACES

A trace representation for benchmark prediction should place two traces close together when they reflect similar execution behavior on a task. Concretely, we consider five attributes of a trace that a representation might encode: i) task (i.e., which task the trace corresponds to), ii) correctness (i.e., whether the trace produced a correct output), iii) system identity (i.e., which system produced the trace), iv) model family (i.e., the family of the underlying generative model), and v) harness (i.e., the harness used to produce the trace). For efficient benchmarking with traces, an ideal representation encodes (i) and (ii) but not (iii)–(v). TRUBRIC constructs such a representation in two stages.

First, for each task q_j , an LLM is prompted with the task description to instantiate a task-specific rubric: a set of K fields $R_j = (r_j^{(1)}, \dots, r_j^{(K)})$, where each field describes one aspect of execution behavior (e.g., how the relevant code was localized). Rubrics are shared across all systems and field names are shared across tasks. Unless otherwise stated, we use $K = 6$ fields.

Second, for each trace x_{ij} we use an LLM to produce a short summary $z_{ij}^{(k)}$ for each field. The summary of each field is then embedded separately with a semantic embedding model $g : \mathcal{X} \rightarrow \mathbb{R}^p$ and concatenated to yield the trace representation $\phi(x_{ij}) = (g(z_{ij}^{(1)}), \dots, g(z_{ij}^{(K)})) \in \mathbb{R}^{K \cdot p}$.

To quantify what a representation encodes, we train class-balanced linear classifiers to predict each of the five attributes of a held-out trace from its embedding. Classifiers for system-level attributes (iii)–(v) are evaluated on held-out tasks. Classifiers for (i) and (ii) are evaluated on held-out groups of systems. We use cached traces from the SWE-bench Verified leaderboard (OpenAI, 2024), with DeepSeek-V4-Flash (DeepSeek-AI, 2026) as both the rubric generator and summarizer. We compare the quality of the TRUBRIC embeddings, using g to embed the start and end of a raw trace, and generic rubric-based embeddings along axes (i)–(v) in Figure 1. Relative to baselines, TRUBRIC preserves (i) and (ii) while mitigating (iii)–(v) across seven modern text encoders. Appendix A provides a complete worked example (i.e., a task description, its TRUBRIC, and extractions for two traces) and more detailed descriptions of the classification tasks.

3.2 INFERENCE IN THE PRODUCT KERNEL PERSPECTIVE SPACE

Given trace representations, we represent systems via the Product Kernel Perspective Space (PKPS). Before constructing the geometry of systems, trace representations are centered by subtracting

task-level means – the average across systems for each task and field – to emphasize differences in behavior. Let τ denote a text embedding of the task description and $\kappa_\sigma(q, q') = \exp(-\|\tau(q) - \tau(q')\|^2 / 2\sigma^2)$ a Gaussian kernel on tasks. For systems f_a and f_b evaluated on task sets Q_a and Q_b , define the weighted similarity

$$\langle a, b \rangle = \frac{\sum_{q \in Q_a} \sum_{q' \in Q_b} \kappa_\sigma(q, q') \langle \phi(x_{aq}), \phi(x_{bq'}) \rangle}{\sum_{q \in Q_a} \sum_{q' \in Q_b} \kappa_\sigma(q, q')}, \quad (1)$$

the corresponding squared distance $\delta^2(a, b) = \langle a, a \rangle + \langle b, b \rangle - 2\langle a, b \rangle$, and $\Delta_{ab} = \delta^2(a, b)$. With Δ defined on $f_0, \dots, f_n$, applying classical multidimensional scaling (MDS) to Δ yields d -dimensional Euclidean representations $(\psi_0, \psi_1, \dots, \psi_n) \in \mathbb{R}^{(n+1) \times d}$ of the systems with respect to their task sets. Importantly, Eq. (1) is defined when $|Q_a| \neq |Q_b|$ if both $|Q_a| > 0$ and $|Q_b| > 0$, so the target system is naturally represented in the same space as the reference systems.

In benchmarking settings, it is typical that $|Q_0| = m \ll M$ and $|Q_i| = M$ for all $i \in \{1, \dots, n\}$. Given $\mathcal{D}$ and $\psi_1, \dots, \psi_n$, we train a simple regressor on $(\psi_1, y_1), \dots, (\psi_n, y_n)$ and apply it to ψ_0 to obtain $\hat{y}_{geom}$. In practice, all hyperparameters – the task kernel bandwidth σ , the dimension d , and the ridge penalty – are selected via cross-validation. Figure 2 shows the $d = 2$ perspectives when using the raw-trace embeddings (top) and the TRUBRIC embeddings (bottom) for different number of references models and number of executed tasks per system. Each dot is a system colored by its full benchmark score on SWE-bench Verified. The TRUBRIC-based PKPS contains more information, per a lower mean absolute error (MAE) on leave-one-system-out evaluation (LOSO; see Section 4).

The geometry-based prediction does not use any item-level score information. To incorporate this information we fit a two-parameter logistic item-response model (Birnbbaum, 1968) with $\mathcal{D}$. We apply the fitted item-response model to f_0 ’s m scored traces to obtain $\hat{y}_{IRT}$. We then consider the combined estimator $\hat{y}_\alpha = \alpha \hat{y}_{IRT} + (1 - \alpha) \hat{y}_{geom}$, with $\alpha \in [0, 1]$ selected via cross-validation. Fitting and prediction details for the item-response model, along with all baselines, are provided in Appendix B.

4 EXPERIMENTS

We next describe the datasets we use to evaluate trace-based geometries for efficient evaluation, introduce three different evaluation protocols relevant to efficient agent evaluation, and describe relevant baselines. We then evaluate TRUBRIC-based estimators against these baselines across benchmarks, evaluation protocols, task budgets, and design choices (Section 4.1).

Benchmark data. We consider two agentic benchmarks:

1. **SWE-bench Verified** (Jimenez et al., 2024; OpenAI, 2024) consists of 500 human-validated software-engineering tasks drawn from real GitHub issues. Its public leaderboard contains 119 submissions; we retain the 107 with complete public trace archives and per-task outcomes. We construct trace representations on a fixed panel of $M = 100$ tasks (a seeded random subset of the tasks with near-universal trace coverage) and take each system’s benchmark score y to be its official resolve rate over all 500 tasks.

2. **Terminal-Bench 2.0** (Terminal-Bench Team, 2026) consists of 89 command-line tasks executed under a standardized harness specification. Its leaderboard contains 75 submissions; we exclude two whose traces are not public and one duplicate submission, and of the remaining 72 we evaluate the 65 with trace coverage of at least 90%. Official scores average five trials per task; we use the traces and binary outcomes of a single trial and take y to be the official all-trial mean.

Evaluation protocols. Modern leaderboards contain many entries that share an underlying model or harness: predicting the score of a system whose near-siblings are among the references is easy – the reference scores can simply be copied – and does not reflect the intended use case of evaluating a genuinely new system. We therefore consider three reference-exclusion protocols of increasing conservatism:

1. *Leave-one-system-out*: only the target itself is excluded from the references.
2. *Leave-one-family-out*: references whose underlying model family matches the target’s are additionally excluded.

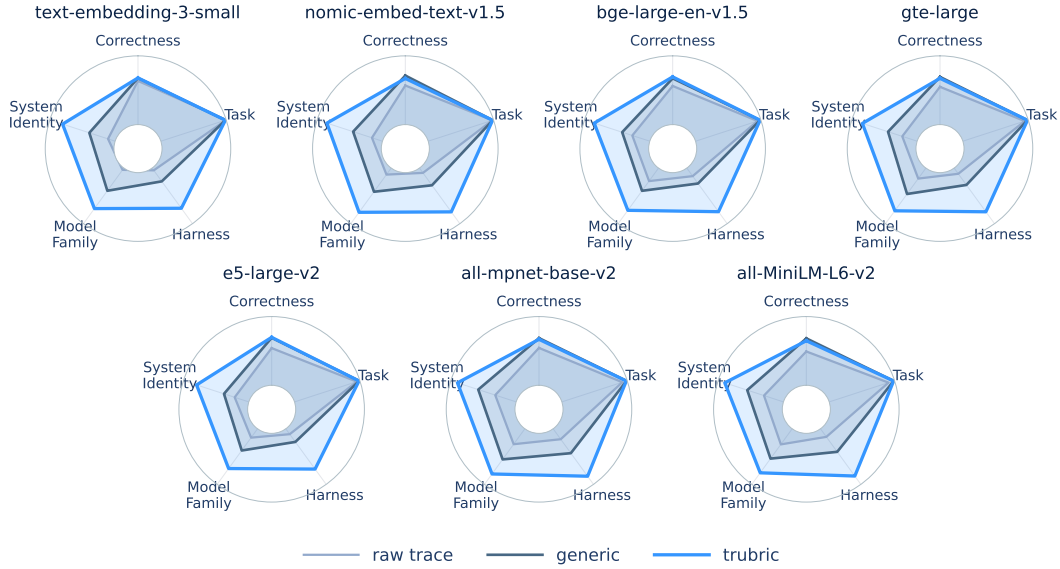


Figure 1: Comparisons of the properties of three different methods for embedding executed task traces across seven different off-the-shelf text encoders: embedding the raw trace, embedding a structured summary based on a generic rubric, and embedding a structured summary based on a task-specific rubric. As measured by the balanced accuracy of held-out linear probes, described in Section 3, the embedding of the task-specific summary is better at removing undesirable signal while preserving relevant signal across all seven encoders.

3. *Leave-one-harness-out*: references sharing the target’s harness are excluded.

Family and harness labels are derived from submission metadata; systems without a label exclude only themselves. Unless stated otherwise, we report leave-one-family-out. For each task budget m , random task selection averages over 50 shared draws; adaptive task selection follows the information-based procedure of Appendix B.

Baselines. We compare against four baselines:

1. *Sample score*: the mean of the target’s m observed trace-level scores.
2. *IRT*: the two-parameter logistic model of Section 3.2.
3. *Raw trace*: the PKPS pipeline applied to embeddings of the start and end of each rendered trace.
4. *Generic*: the PKPS pipeline applied to summaries produced with fixed, task-agnostic field descriptions.

Implementation details for all baselines are provided in Appendix B.

4.1 RESULTS

Rubric-based preprocessing produces more useful trace representations. We compare embeddings of raw traces with embeddings of structured summaries produced using either a generic rubric or a task-specific rubric, TRUBRIC. Figure 1 shows that both rubric constructions reduce recoverable information about system identity, model family, and harness while retaining task and correctness information. As discussed in Section 1, these attributes can introduce nuisance variation into trace similarity, obscuring similarities in how systems execute tasks. Task-specific rubrics generally produce a larger reduction in identity-related signals than generic rubrics. This qualitative pattern persists across seven embedding functions with little overall variation, although the precise magnitudes differ.

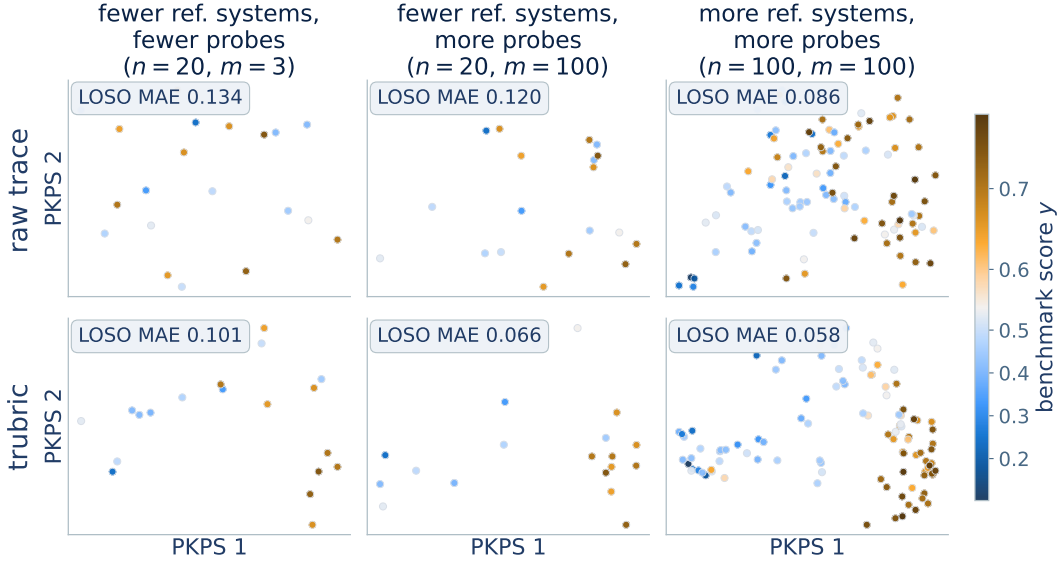


Figure 2: The 2-d Product Kernel Perspective Space (PKPS) induced with raw trace embeddings (top) and the TRUBRIC embeddings (bottom) under different number of reference systems and number of task executions. Each dot is a system colored by full benchmark score on SWE-bench Verified. The manifold of systems under the TRUBRIC embeddings is more strongly correlated with the full benchmark score in paired settings. We report the average Mean Absolute Error (MAE) under leave-one-system-out (LOSO) protocol to quantify this observation.

These differences are also reflected in the system-level geometry. Figure 2 shows a clearer relationship between system position in the embedding space and benchmark score for TRUBRIC than for raw-trace embeddings. In Figure 3, prediction error decreases more rapidly with additional tasks when using TRUBRIC representations than when using raw-trace embeddings. On SWE-bench Verified, increasing the task budget from 1 to 20 reduces geometry-only MAE from .081 to .060 for TRUBRIC, compared with .099 to .089 for raw traces. Table 1 further shows that generic rubrics are competitive on SWE-bench, particularly at small budgets, though task-specific rubrics provide larger improvements on Terminal-Bench 2.0.

Trace geometry alone supports task-efficient benchmark prediction. Table 1 and Figure 3 show that rubric-based trace geometry predicts full-benchmark performance more accurately than the sample score across small task budgets. Under leave-one-family-out evaluation, this advantage holds from 1 through 10 randomly selected tasks on both benchmarks. The geometric estimator also outperforms the population-mean baseline, indicating that its accuracy reflects useful differences between systems rather than just regularizing towards the average reference score. We emphasize that these geometric methods *only* require the target’s trace content, and *not* its correctness labels, which means that these methods are applicable in settings where grading (even a small number of) target tasks is infeasible (due to e.g. cost, lack of expertise, etc.), provided scored reference systems are available.

Trace geometry can also outperform IRT, which already exploits the correctness patterns of reference systems. On SWE-bench, geometry remains more accurate through 10 random tasks (Figure 3). On Terminal-Bench, geometry outperforms IRT at one task, but IRT is more accurate from three tasks onward. The advantage of trace geometry over outcome-only methods is largest when few target executions are available, while its relative performance at larger budgets varies by benchmark.

Combining trace geometry with correctness outcomes improves prediction beyond either source alone. The two estimators exhibit different behavior as the task budget grows: geometric predictions are informative with very few executions, while IRT improves substantially as the number of tasks increases. We combine the geometric and IRT predictions using the weighting procedure described in Section 3, referring to the resulting estimator as IRT + TRUBRIC. Figure 3

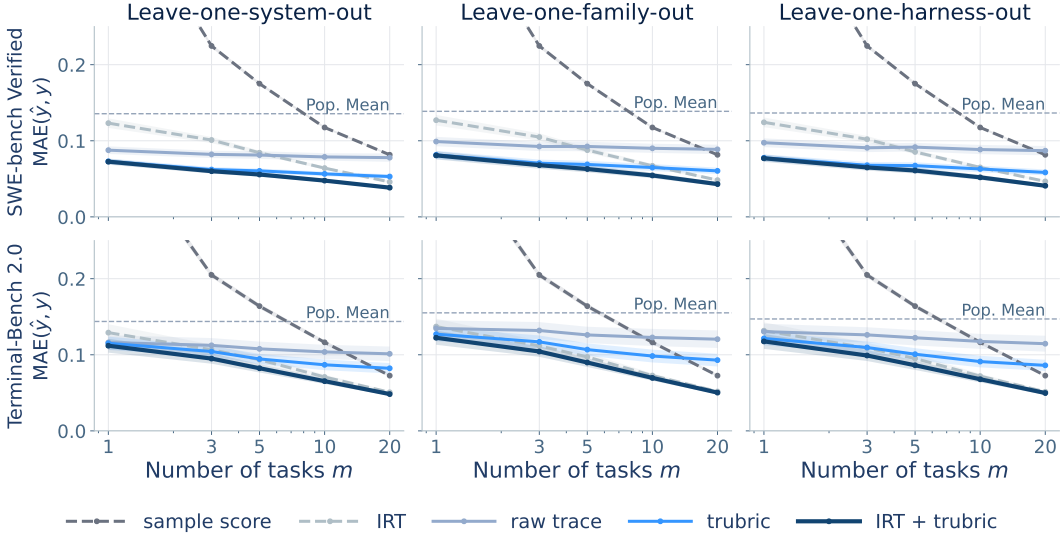


Figure 3: The Mean Absolute Error (MAE) of benchmark score estimators as a function of the number of executed tasks for SWE-bench Verified (top) and Terminal-Bench 2.0 (bottom) under three different realistic evaluation protocols: leave-one-system-out (left), leave-one-family-out (center), and leave-one-harness-out (right). Reported MAE is the average over 50 random subsets of tasks. Shaded areas represent ± 1 S.E. The TRUBRIC and IRT ensemble dominates all methods (i.e., is never worse and is sometimes better) across all protocols, datasets, and number of executed tasks.

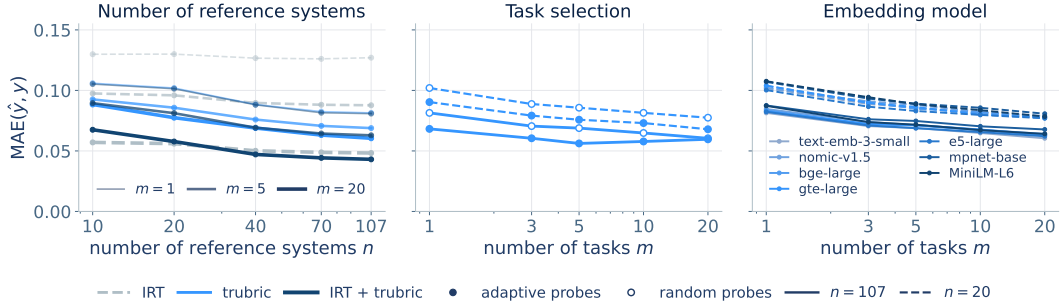


Figure 4: Mean Absolute Error (MAE) of benchmark score estimators with access to different number of reference systems (left), different task selection procedures (center), and using different text encoders (right) under the leave-one-family-out evaluation protocol. Reported MAE is the average over 50 random subsets of tasks; adaptive task paths are deterministic given the reference systems. Adaptive tasks are selected sequentially to maximize Fisher information under a two-parameter logistic model fit to the reference systems, as described in Section 3.

shows that combining their predictions improves accuracy across evaluated budgets. At five random tasks on SWE-bench, IRT + TRUBRIC achieves .063 MAE, compared with .069 for geometry alone and .088 for IRT. On Terminal-Bench – where IRT is relatively stronger – the combination achieves .090, compared with .107 for geometry and .097 for IRT.

The improvement therefore extends beyond choosing whichever method performs better on a particular benchmark. These results support the complementary predictive value of execution content and correctness outcomes. The broad pattern persists across the system-, family-, and harness-exclusion protocols in Figure 3, although the size of the improvement varies. As Table 1 shows, IRT + TRUBRIC is best or near-best across the reported settings, though its advantage over IRT generally shrinks as the number of correctness observations increases.

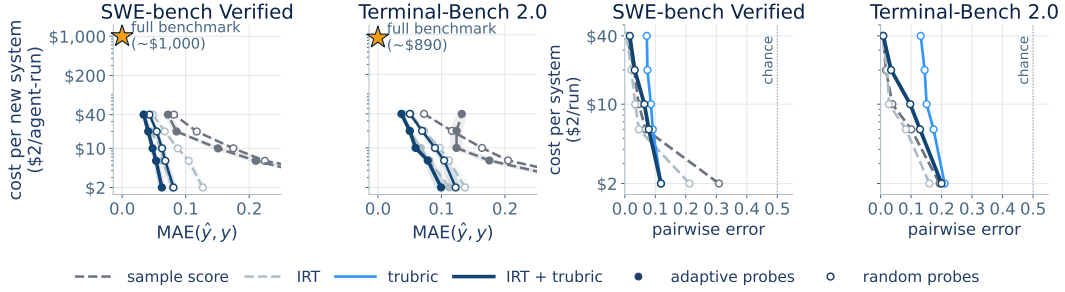


Figure 5: The cost of achieving a particular Mean Absolute Error (MAE; left and center left) and pairwise ranking error (center right and right) when using different benchmark score estimators on SWE-bench Verified and Terminal-Bench 2.0. Stars indicate the cost of producing the true benchmark score. An ideal method has low error (to the left) and is cheap (lower). Costs are calculated at a nominal \$2 per task execution. Values are estimated using the leave-one-family-out results underlying Table 1 and, for pairwise error, leave-two-out comparisons of system pairs whose true scores differ by at least 0.05.

Trace-informed estimation reduces the cost of reaching a given prediction accuracy. Figure 5 translates the error curves into execution costs, illustrating how more informative estimates can reduce the number of tasks required for a desired accuracy. For this comparison, we assume a cost of \$2 per agent execution. On SWE-bench, IRT + TRUBRIC reaches an MAE below .10 with one random task, compared with five tasks for IRT and twenty for the sample score – the smallest tested budgets meeting this threshold. This corresponds to estimated target execution costs of \$2, \$10, and \$40, respectively.

Combining trace information with adaptive sampling also reduces cost at a more stringent accuracy level: five adaptive tasks with IRT + TRUBRIC achieve approximately the same error as twenty random tasks with IRT, .048 MAE (Table 1), corresponding to \$10 rather than \$40. These comparisons concern the cost of executing tasks for a new system and assume an existing reference collection. We assume that reference executions are available from community archives or that their acquisition cost is amortized over a large number of target evaluations. In this calculation, we treat the cost of LLMs used to do (t)rubric summarization and the cost of API-based embedding functions as negligible compared to the cost of running the agents on the task. These comparisons illustrate how extracting additional information from each execution can make accurate benchmark estimation feasible at substantially smaller task budgets.

Performance improves with larger reference libraries and adaptive task selection. Figure 4a shows that increasing the number of reference models generally improves both IRT and geometry-based estimators. At five random tasks on SWE-bench, expanding the library from 20 to 107 systems reduces geometry-only MAE from .086 to .069 and IRT + TRUBRIC MAE from .081 to .063. The contribution of geometry depends on reference availability: given a small number of reference models and a large target-task budget, IRT can outperform geometry-based methods.

Adaptive task selection provides a further improvement. With five adaptive tasks, IRT + TRUBRIC achieves .048 MAE on SWE-bench, compared with .063 for five random tasks (Table 1). Together, Table 1 and Figure 4b show that adaptive selection improves both IRT and geometry-based prediction. These findings demonstrate that trace-based estimation can be used alongside adaptive sampling.

Figure 4c further shows that the qualitative behavior persists across seven embedding functions. On SWE-bench, the impact of encoder choice appears small compared with the effects of the number of target tasks and the number of reference models.

Broader rubrics improve average performance, though strong, small rubrics may exist. Figure 6 varies the number of fields included in generic and task-specific rubrics, averaging performance over subsets drawn from a bank of 32 fields. For both constructions, adding fields generally improves average prediction accuracy, with diminishing returns. At five tasks and the larger refer-

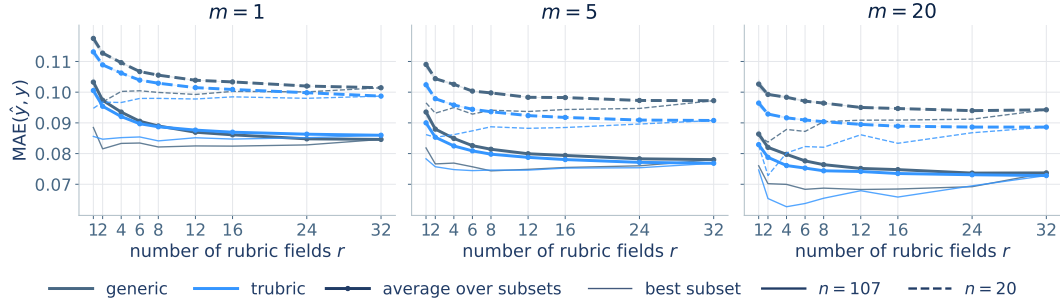


Figure 6: The effect of the number of fields present in the rubric used to extract a structured summary of the trace on Mean Absolute Error (MAE) for different number of executed tasks and different number of reference systems for both task-specific and generic rubrics. We report both the MAE for the average (bold) and best performing (thin) sets of fields. In general, a task-specific rubric is preferred to a generic rubric when there are fewer reference systems and more executed tasks available.

ence library, increasing a task-specific rubric from one to six fields reduces average MAE from .090 to .081; using all 32 fields reduces it further to .077. This pattern is consistent with an ensembling effect from combining descriptions of multiple aspects of execution. Task-specific rubrics generally outperform generic rubrics in this experiment, with small reversals at the one-task budget.

The thin curves show that some small subsets perform substantially better than the average subset of the same size. Each point selects the lowest average evaluation error among the tested subsets separately for that setting, averaging over probe draws and all 107 target systems even when the reference library contains 20 systems. These retrospective selections demonstrate the existence of favorable small rubrics on the evaluation panel, rather than a validated procedure for identifying them in advance. The selected subsets need not be nested as field count increases, so their performance need not improve monotonically.

5 DISCUSSION

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A TRACE REPRESENTATION DETAILS

A.1 EXAMPLE TASK, TRUBRIC, AND EXTRACTIONS

We illustrate the TRUBRIC pipeline on SWE-bench Verified instance `django__django-11099`.

Task. The problem statement, as provided to the rubric-instantiation LLM:

UsernameValidator allows trailing newline in usernames
Description

ASCIIUsernameValidator and UnicodeUsernameValidator use the regex

`r'^[w. @+ ]+$'`

The intent is to only allow alphanumeric characters as well as `.`, `@`, `+`, and `-`. However, a little known quirk of Python regexes is that `$` will also match a trailing newline. Therefore, the user name validators will accept usernames which end with a newline. You can avoid this behavior by instead using `\A` and `\Z` to terminate regexes. For example, the validator regex could be changed to

`r'^\A[w. @+ ]+\Z'`

in order to reject usernames that end with a newline.

I am not sure how to officially post a patch, but the required change is trivial - using the regex above in the two validators in `contrib.auth.validators`.

Instantiated TRUBRIC. The six task-specific field descriptions generated for this instance:

understanding Look for the agent to identify that the issue lies in the regex anchoring with `$` in `ASCIIUsernameValidator` and `UnicodeUsernameValidator`, specifically that `$` matches a trailing newline in Python, and that changing to `\A` and `\Z` is the proposed fix in `django/contrib/auth/validators.py`.

localization Check that the agent correctly locates the two validator classes (`ASCIIUsernameValidator` and `UnicodeUsernameValidator`) in `django/contrib/auth/validators.py` and references the existing regex `r'^[w. @+ ]+$'` in both.

reproduction Look for the agent to write or run a test that confirms a username ending with a newline (e.g., `'validuser\n'`) is currently accepted by the validators, possibly by using Django's `validate_username` or directly instantiating the validators.

editing Verify that the agent modifies the regex pattern in both validator classes from `r'^[w. @+ ]+$'` to `r'^\A[w. @+ ]+\Z'`, ensuring no other changes are made to the file.

verification The agent should run the relevant test suite (e.g., `tests/auth_tests/test_validators.py` or full Django auth tests) to confirm that usernames with trailing newlines are now rejected, and that existing valid usernames still pass.

final state The final state should show `django/contrib/auth/validators.py` with both regex patterns updated, and no other unintended modifications, with tests confirming the newline rejection behavior.

Extractions. Structured summaries of two systems’ traces on this task, produced against the rubric above. The summaries describe execution behavior without reference to the harness or underlying model that produced the trace.

SYSTEM A.

understanding The agent identified that the issue stems from Python’s regex behavior where `$` matches a trailing newline, and that changing the anchors to `\A` and `\Z` in both validators would fix it. It explicitly mentioned this during the initial inspection of the validator file.

localization The agent searched for `validators.py`, found `django/contrib/auth/validators.py`, opened it, and located both `ASCIIUsernameValidator` and `UnicodeUsernameValidator` classes. It acknowledged the existing regex pattern `r'^[w.@+-]+$'` in both classes before making changes.

reproduction The agent attempted to reproduce the bug by creating `test_validators.py` and `simple_test_validators.py`, but these scripts failed due to missing Django settings configuration. It then created `regex_test.py` to directly test the regex patterns, which ran successfully and confirmed rejection of a trailing newline after the fix, though it did not demonstrate acceptance before the fix.

editing The agent edited lines 10 and 20 in `django/contrib/auth/validators.py`, changing the regex from `r'^[w.@+-]+$'` to `r'\A[w.@+-]+\Z'` in both `ASCIIUsernameValidator` and `UnicodeUsernameValidator`. The diff shows only these two regex changes to the file, with no other modifications.

verification The agent ran the custom `regex_test.py` script, which output that both `ASCII` and `Unicode` validators rejected a username ending with a newline. It did not run the existing Django test suite (e.g., `tests/auth_tests/test_validators.py`) to verify the fix against formal tests.

final state The final state includes `django/contrib/auth/validators.py` with both regex patterns updated to `r'\A[w.@+-]+\Z'`. Three new files (`test_validators.py`, `simple_test_validators.py`, `regex_test.py`) were created, but no other existing files were modified. The agent submitted these changes.

SYSTEM B.

understanding The agent identified that the issue stems from Python’s regex behavior where `$` matches a trailing newline, causing `ASCIIUsernameValidator` and `UnicodeUsernameValidator` to incorrectly accept usernames ending with newlines. It understood that replacing `^` and `$` with `\A` and `\Z` would fix the anchoring issue, as described in the issue for `django/contrib/auth/validators.py`.

localization The agent located the two validator classes, `ASCIIUsernameValidator` and `UnicodeUsernameValidator`, in `django/contrib/auth/validators.py`. It confirmed both classes use the existing regex pattern `r'^[w.@+-]+$'` and identified these as the exact locations requiring modification.

reproduction The agent verified the issue by running a quick test comparing the old and new regex patterns against usernames with and without trailing newlines. The test confirmed that the old pattern matched usernames ending with a newline, while the new pattern correctly rejected them, and valid usernames still passed.

editing The agent modified the regex pattern in both validator classes from `r'^[w.@+-]+$'` to `r'\A[w.@+-]+\Z'` in `django/contrib/auth/validators.py`. The diff shows only these two regex changes, with no other modifications made to the file.

verification The agent ran the auth test suite which passed all 494 tests, including the tests in `auth_tests/test_validators.py` for `UsernameValidatorsTests`. It also ran a targeted test confirming that usernames with trailing newlines are now rejected by both validators while valid usernames still pass.

final state The final state shows `django/contrib/auth/validators.py` with both regex patterns updated to `r'\A[w.@+-]+\Z'` in `ASCIIUsernameValidator` and `UnicodeUsernameValidator`. No other files were modified, and tests confirm the newline rejection behavior works correctly.

A.2 LINEAR-PROBE METHODOLOGY

For each combination of text encoder and trace representation, we assemble one embedding per trace for the executions of the 107 SWE-bench Verified systems on a fixed panel of 20 tasks (2,129 traces; a small number of executions are missing from the public archive). Rubric-based representations are the concatenated per-field embeddings of Section 3.1; raw-trace representations embed head and tail slices of the rendered log as two blocks. Each block is ℓ_2 -normalized and blocks are weighted equally.

Table 2: Held-out balanced accuracy of linear probes for the five trace attributes of Section 3.1, for each text encoder and trace representation; these values underlie Figure 1. For evaluation-relevant attributes (i)–(ii), higher is better; for system-metadata attributes (iii)–(v), lower is better. The final row reports chance balanced accuracy.

encoder	representation	(i) task	(ii) correctness	(iii) system	(iv) family	(v) harness
text-embedding-3-small	raw trace	1.00	0.82	0.89	0.97	0.96
	generic	0.99	0.84	0.61	0.64	0.76
	TRUBRIC	0.99	0.84	0.20	0.36	0.29
nomic-embed-text-v1.5	raw trace	0.98	0.78	0.84	0.90	0.91
	generic	0.99	0.86	0.56	0.63	0.70
	TRUBRIC	0.99	0.84	0.16	0.30	0.22
bge-large-en-v1.5	raw trace	0.95	0.78	0.73	0.79	0.86
	generic	0.99	0.84	0.58	0.64	0.73
	TRUBRIC	0.99	0.85	0.16	0.33	0.23
gte-large	raw trace	0.95	0.77	0.77	0.83	0.90
	generic	0.99	0.85	0.56	0.59	0.70
	TRUBRIC	0.99	0.84	0.17	0.33	0.23
e5-large-v2	raw trace	0.96	0.77	0.79	0.86	0.90
	generic	0.99	0.85	0.62	0.66	0.77
	TRUBRIC	0.99	0.85	0.21	0.37	0.29
all-mpnet-base-v2	raw trace	0.93	0.77	0.68	0.76	0.82
	generic	0.99	0.84	0.43	0.52	0.57
	TRUBRIC	0.99	0.83	0.11	0.29	0.16
all-MiniLM-L6-v2	raw trace	0.91	0.75	0.70	0.75	0.86
	generic	0.99	0.84	0.45	0.53	0.59
	TRUBRIC	0.99	0.82	0.11	0.31	0.16
chance		0.05	0.50	0.01	0.12	0.02

Probes are linear classifiers trained with class-balanced squared loss and a ridge penalty; the penalty is selected along a closed-form regularization path using grouped cross-validation within the training folds. Evaluation is five-fold grouped cross-validation with seeded, size-balanced group allocation: for the system-level attributes (iii)–(v) the groups are tasks, so probes are evaluated on tasks unseen during training, and for (i)–(ii) the groups are model families, so probes are evaluated on systems from families unseen during training. We report balanced accuracy averaged over folds; chance balanced accuracy for a C -class attribute is $1/C$. Figure 1 plots $(\text{accuracy} - \text{chance}) / (1 - \text{chance})$ for attributes (i)–(ii) and its complement for (iii)–(v), so that better-for-evaluation is always outward. Table 2 reports the balanced accuracies underlying Figure 1.

A.3 PROMPTS

The rubric-instantiation prompt (`{problem}` is the task description):

Here is a repository issue that a software-engineering agent will be asked to fix:

`{problem}`

Write an instance-specific rubric for judging an agent’s trace on THIS issue: for each section below, one sentence saying what specifically to look for in this instance (name the likely relevant modules/files/tests if inferable).

Sections: understanding, localization, reproduction, editing, verification, final_state
Respond with ONLY a JSON object with those keys, each a one-sentence string.

The extraction prompt (`{rubric}` is the instantiated TRUBRIC for the task, or the fixed generic rubric below for the generic baseline; `{keys}` the field names; the rendered trace is appended after the prompt):

Below is the (truncated) execution trace of a software-engineering agent.

Extract a factual description of the agent’s behavior for EACH of the following sections. 30-60 words per section. Describe behavior only -- do not restate the problem. If a section did not occur, write exactly what is missing (e.g. "never ran any test after editing").

{rubric}

Respond with ONLY a JSON object with keys {keys}, each a string.

The generic rubric:

- understanding: how the agent oriented itself before acting
- localization: how it searched/navigated to find the relevant code
- reproduction: whether/how it reproduced the issue before fixing
- editing: what it changed (files, nature and size of the edit)
- verification: whether it re-ran tests or a repro script after editing, and the OBSERVATION
- final_state: how the run ended (clean finish, step/cost limit, submitted unverified)

B EFFICIENT BENCHMARKING DETAILS

B.1 SCORE-ONLY BASELINES

Sample score. The mean of the target’s m observed trace-level scores, $\hat{y}_{\text{sample}} = \frac{1}{m} \sum_{q \in Q_0} s(f_0(q))$.

Population mean. The mean benchmark score of the reference systems, $\hat{y}_{\text{pop}} = \frac{1}{n} \sum_{i=1}^n y_i$, which ignores the target’s executions entirely.

Item-response model. We fit the two-parameter logistic model $\mathbb{P}(y_{ij} = 1) = \text{sigmoid}(a_j(\theta_i - b_j))$ to the reference score matrix by maximum a posteriori estimation, with standard normal priors on the abilities θ_i and difficulties b_j and a $\mathcal{N}(0, 1/10)$ prior on the log-discriminations $\log a_j$ (centering the slopes at the Rasch value of one), optimized with L-BFGS. Because the benchmark score aggregates all M tasks while the model is fit on the M -task panel, we additionally fit a monotone link from ability to benchmark score, $y \approx \text{sigmoid}(c_1\theta + c_0)$, by least squares on the reference systems. For a target, we form a posterior over ability on a dense grid – the prior is a normal with the mean and standard deviation of the reference abilities – update it with the likelihood of the target’s m observed trace-level scores, and report the posterior mean of the fitted ability-to-score link as $\hat{y}_{\text{IRT}}$.

Adaptive task selection. Given the fitted item-response model, tasks are selected greedily per target: at each step, the next task is the one with the highest expected Fisher information $a_j^2 p_j(\theta)(1 - p_j(\theta))$ under the current ability posterior, the outcome is observed, and the posterior is updated. Random task selection instead draws Q_0 uniformly; reported random-selection results average over 50 shared draws.

B.2 TRACE-GEOMETRY BASELINES

The raw-trace and generic-rubric estimators replace the TRUBRIC representation ϕ in the pipeline of Section 3.2 and are otherwise identical. The raw-trace representation embeds the first and last portions of the rendered execution log as two blocks with the same encoder g ; representations are centered by the task-level median across systems and ℓ_2 -normalized. The generic-rubric representation follows the two-stage construction of Section 3.1 with the fixed field descriptions of Appendix A.3 in place of task-specific rubrics.

B.3 HYPERPARAMETERS AND CROSS-VALIDATION

For each task draw, hyperparameters are selected by treating each reference system as a pseudo-target: its score is predicted from the remaining references (leave-one-out via the hat matrix for

the ridge regressor), and the configuration minimizing the mean reference error is selected. The candidate grids are the kernel bandwidth $\sigma^2 \in \{\text{med}/2^k\}_{k=0}^8$ where med is the median squared distance between task embeddings, the dimension $d \in \{8, 16\}$, and the ridge penalty $\in \{0.1, 1.0\}$. The ensemble weight α is selected on a 101-point grid over $[0, 1]$ by the same reference-pseudo-target procedure.